ARSHID AHMAD MALIK

Electronics & Communication Engineer | EV Systems | Embedded & IoT | EV Charging Infrastructure
Bengaluru, Karnataka, India | +91 9682397579 | malikarshid01430@gmail.com |

PROFESSIONAL SUMMARY

Electronics & Communication Engineering graduate with hands-on experience across embedded systems, IoT, VLSI Design for Testability, and EV Charging Infrastructure, including an industry-sponsored EV training program backed by Mercedes-Benz R&D India (MBRDI) and TERI. Skilled in firmware development, diagnostic fault-finding, and technical documentation, with practical exposure to Battery Management Systems, AC/DC charging standards, and embedded hardware debugging. Seeking to contribute to EV design, product development, testing, validation, or technical support roles in the electric vehicle and automotive industry.

TECHNICAL SKILLS

- Programming Languages: C, C++, Embedded C, Verilog HDL, Kotlin
- Microcontrollers & Platforms: ESP32, ESP8266, STM32 (ARM Cortex-M), Arduino UNO/Nano, Node-MCU, 8051 basics
- Embedded Systems & Firmware: Firmware Development & Debugging, Interrupts/Timers, GPIO, PWM, ADC/DAC
- Communication Protocols: UART, SPI, I2C, MQTT, Wi-Fi, TCP/IP, HTTP/HTTPS
- EV & Automotive Systems: Battery Management Systems (BMS), EV Charging Infrastructure (AC/DC Standards), Electric Powertrain Fundamentals, Vehicle Architecture, EV Safety Standards, Vehicle Diagnostics
- Hardware, PCB & Tools: Circuit & PCB Design, Oscilloscope, Multimeter, Git/GitHub, Testing & Quality Assurance

EXPERIENCE

EV Charging Infrastructure Skilling Trainee - "Future in Charge" Program - VisionAstraa EV Academy | *Jul 2026 - Aug 2026*
Sponsored by Mercedes-Benz Research and Development India (MBRDI) and TERI

- Completed industry-sponsored training in EV charging infrastructure design and commissioning, diagnosing faults on AC/DC charging equipment and studying BMS integration under standard safety and compliance protocols.

IoT Systems Intern - Emertex | *Mar 2026 - Apr 2026*

- Designed and debugged ESP32/Arduino-based IoT systems with a multi-sensor MQTT cloud pipeline, achieving sub-2-second diagnostic-to-resolution turnaround using oscilloscopes and serial monitors.

Android App Development Intern - Mind Matrix Inc. | *Feb 2026 - May 2026*

- Developed Kotlin Android applications integrating the Google Gemini Pro API via Jetpack Compose, reducing response latency by 25% through optimized prompt-chaining and request-handling logic.

VLSI Design for Testability (DFT) Intern - VLSIGuru | *Feb 2026 - May 2026*

- Analyzed scan chain architectures and stuck-at fault-detection methods, validating chip-level designs using ATPG concepts, boundary scan, and DFT insertion flows.

Artificial Intelligence Intern - CODEC Technologies | *Aug 2025 - Sep 2025*

- Trained ML classification models with 15% accuracy improvement over baseline and automated preprocessing on 5,000+ record datasets, cutting manual effort by 30% through Python-based pipelines.

PROJECTS

- **Solar-Powered Smart Indoor Air Purifier** - Diagnosed and optimized ESP32-based air-quality monitoring system, cutting active power use by 20% through deep-sleep tuning and MQTT-based real-time data streaming.
- **IoT-Based Smart Home Automation System** - Deployed and maintained a multi-node ESP32 control system for 6+ appliances, achieving sub-1.5s response time and improving system stability by 25% via watchdog-based error recovery.
- **Aircraft Anti-Collision Monitoring System** - Developed an interrupt-driven Arduino system with sub-100ms alert response and UART telemetry logging, validated across 50+ simulated scenarios.
- **Obstacle-Avoiding Robot with Lane Detection** - Programmed autonomous navigation on Arduino UNO using IR and ultrasonic sensors, improving obstacle-detection accuracy by 30% through iterative sensor calibration.

EDUCATION

Bachelor of Engineering - Electronics & Communication Engineering | MVJ College of Engineering (VTU), Bengaluru | 2023 – 2026 | CGPA: 6.5
Diploma - Electronics & Communication Engineering | Government Polytechnic College, Srinagar (JKBOTE) | 2020 – 2023 | GPA: 71%
PUC (PCMB) | Govt. Model Higher Secondary School, Dooru (JKBOSE) | 2018 – 2020 | GPA: 58%
SSLC (Science) | Army Goodwill School, Qazigund (JKBOSE) | 2017 | GPA: 59%

CERTIFICATIONS

NPTEL - Microsensors & Nanosensors, Classics of Neuroscience, Cloud Computing | Arduino Training Certification | Introduction to IoT | Oracle Certified Foundations Associate | Hexart Certificate of AI Foundation and AI Advanced | Introduction to Data Analytics | Ethical Hacking for Beginners | Cybersecurity

LANGUAGES

English, Hindi, Urdu, Kashmiri